

Highlights

What Do LLMs Say That Human Facilitators Don't? A Computational Behavioral Analysis of AI vs. Human Collaborative Design Meeting Facilitation

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- LLMs and humans differ significantly on 4 of 6 facilitation behavior dimensions
- Human facilitators produce more semantically novel facilitation moves
- LLMs generate topically divergent responses that depart further from prior context
- LLM facilitation behavior is largely temperature-stable; only specificity and phase appropriateness show modest temperature sensitivity
- Findings inform the behavioral design of AI facilitation tools

What Do LLMs Say That Human Facilitators Don't?

A Computational Behavioral Analysis of AI vs. Human Collaborative Design Meeting Facilitation

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Abstract

Artificial intelligence is increasingly proposed as a tool for facilitating collaborative design meetings, yet no systematic computational comparison exists between LLM-generated and human facilitation behaviors. This study addresses this gap through a quantitative behavioral analysis of 199 matched facilitation move pairs drawn from the AMI Meeting Corpus (Carletta et al., 2005), comparing trained human facilitators (Project Managers) with Llama 3.1 8B Instruct (Dubey et al., 2024) across six behavioral dimensions: semantic novelty, directiveness, specificity, empathy, topic divergence, and phase appropriateness. Each dimension was computed using validated NLP methods including sentence-transformer embeddings (Reimers and Gurevych, 2019), a fine-tuned DistilBERT classifier (Devlin et al., 2019), and LDA-based topic modeling (Blei et al., 2003). Inter-rater reliability for facilitation move identification was $\kappa = 0.81$. Intra-facilitator consistency analysis across sessions confirmed high behavioral consistency for novelty (ICC = 0.78) and moderate consistency for divergence (ICC = 0.22), supporting the validity of treating PM turns as representative of individual facilitation style rather than session-specific noise. Mann-Whitney U tests with Bonferroni correction revealed significant differences on four of six dimensions ($p < 0.05$, 12/18 comparisons). Human facilitators produced significantly more semantically novel responses ($r = -0.54$), while LLMs exhibited greater topic divergence ($r = +0.58$), higher specificity ($r = +0.25$), and more phase-appropriate terminology ($r = +0.34$). No statistically significant difference was detected for

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directiveness or empathy; both dimensions were underpowered (power = 0.05 and 0.21 respectively), and these null results should not be interpreted as evidence of equivalence. Temperature variation (0.3–1.0) significantly affected specificity ($\rho = +0.12$, $p = 0.005$) and phase appropriateness ($\rho = -0.10$, $p = 0.013$); the remaining four dimensions were temperature-stable. These findings suggest that current LLMs adopt a characteristically different facilitation style from human experts, with implications for the design of AI-augmented facilitation tools.

Keywords: collaborative design meetings, facilitation, large language models, behavioral analysis, human-AI collaboration, AMI Meeting Corpus

1. Introduction

Collaborative design meetings are a primary site of creative problem-solving in industry, education, and public policy (Brown, 2008; Cross, 2011). Central to these collaborative processes is the role of the facilitator, whose
5 verbal interventions—questions, reframings, provocations—shape how teams explore problem and solution spaces (Seidel and Fixson, 2013; Starostka et al., 2021). As large language models (LLMs) become increasingly capable of generating contextually appropriate and coherent text, researchers and practitioners have begun exploring their potential as AI-powered facilitation
10 assistants across online discussion (Korre et al., 2025) and collaborative design meeting contexts (Gmeiner et al., 2023). Yet a fundamental empirical question remains unanswered: how do the facilitation behaviors of LLMs differ from those of trained human facilitators? Industry practitioners at design consultancies and educators at engineering and business schools have begun
15 piloting LLM-driven facilitation prompts in live design sessions (Gmeiner et al., 2023), yet without evidence-based guidance on how AI facilitation behavior actually differs from expert human practice.

This question is critical for the behavioral sciences because facilitation moves are not merely linguistic acts. They are observable behavioral units
20 that reflect underlying communicative goals, mental models of the design process, and theories of learning and collaboration. Schön (1983) conceptualized professional practice as a form of “reflection-in-action,” in which facilitators draw on embodied repertoires of moves shaped through experience. From a Social Cognitive Theory perspective (Bandura, 1986), human facilitators
25 develop these repertoires through cycles of observation, feedback, social

modeling, and embodied practice. This developmental pathway differs fundamentally from LLM text generation, which produces linguistically plausible outputs through statistical pattern completion over training corpora rather than through situated social learning. The six behavioral dimensions measured in this study—novelty, directiveness, specificity, empathy, divergence, and phase appropriateness—can therefore be treated as computational proxies for facilitation behavior. The AMI Meeting Corpus provides naturalistic recordings of collaborative product design meetings, constituting a direct empirical base for studying facilitation behavior in design contexts.

Despite growing interest in AI-augmented creativity tools (Frich et al., 2019; Dove et al., 2017), the empirical literature lacks a direct, quantitative comparison of human and LLM facilitation behavior within naturalistic design discourse. Prior work on computational analysis of design discourse has focused on coding schemes and protocol analysis (Kan and Gero, 2017; Stempfle and Badke-Schaub, 2002; Chandrasegaran et al., 2019), while studies of LLMs in creative contexts have examined writing assistance (Lee et al., 2022; Yuan et al., 2022) and ideation support (Peng et al., 2025) without isolating the specific behavioral profile of facilitation moves.

This study addresses three research questions:

- **RQ1:** Do LLM-generated and human facilitation moves differ significantly on the six behavioral dimensions?
- **RQ2:** Which dimensions show the largest behavioral differences, and in what direction?
- **RQ3:** How does decoding temperature affect LLM facilitation behavior?

We operationalize these questions through a computational pipeline that extracts facilitation moves from the AMI Meeting Corpus (Carletta et al., 2005), generates matched LLM responses using Llama 3.1 8B Instruct (Dubey et al., 2024) at three temperature settings, and quantifies behavioral differences across six validated dimensions using nonparametric statistical tests.

2. Literature Review

2.1. Collaborative Design Meeting Facilitation

Collaborative design practice is a human-centered approach to innovation structured around iterative phases of empathizing, defining, ideating,

60 prototyping, and testing (Brown, 2008). The facilitator plays a pivotal role in guiding teams through these phases, shaping discourse through questions, provocations, and reframings rather than direct instruction (Seidel and Fixson, 2013). Starostka et al. (2021) developed a taxonomy of facilitation moves that distinguishes between directive moves (e.g., “Let’s move on to
65 prototyping”) and generative moves (e.g., “What if we approached this from the user’s perspective?”). Dorst and Cross (2001) showed that creative outcomes in design depend on the co-evolution of problem and solution frames, a process that skilled facilitators actively manage through targeted verbal interventions. The behavioral repertoire of effective facilitators is thus com-
70 plex, context-sensitive, and developed through extended professional practice (Schön, 1983).

2.2. Computational Analysis of Design Discourse

Computational approaches to design discourse have evolved from manual protocol analysis (Kan and Gero, 2017) toward automated methods using
75 natural language processing. Stempfle and Badke-Schaub (2002) established foundational categories for analyzing team communication in design, while Chandrasegaran et al. (2019) introduced real-time visualization of verbal design content. Semantic similarity metrics based on sentence embeddings (Reimers and Gurevych, 2019) enable automated measurement of novelty and
80 divergence in design conversations. Topic modeling approaches, particularly Latent Dirichlet Allocation (Blei et al., 2003), have been applied to identify thematic structure in design corpora. These computational methods provide the infrastructure for scaling behavioral analysis beyond what manual coding can achieve.

85 2.3. LLMs in Creative and Collaborative Contexts

The emergence of instruction-tuned LLMs has opened new possibilities for AI-augmented creativity (Frich et al., 2019). Lee et al. (2022) demonstrated that LLMs can serve as effective co-writing partners, while Yuan et al. (2022) explored story writing with language models. In the design domain, Peng
90 et al. (2025) reviewed AI-augmented design processes, finding that LLMs excel at generating diverse ideas but lack the contextual sensitivity of human collaborators. Korre et al. (2025) surveyed how LLMs are used for evaluation and facilitation of online discussions, identifying key gaps in behavioral characterization. The Llama 3 family of models (Dubey et al., 2024) represents

95 the current state of the art in open-weight instruction-following LLMs, with
8B-parameter variants suitable for real-time facilitation applications.

2.4. *Human-AI Collaboration: Behavioral Perspectives*

Yang et al. (2020) argued that human-AI interaction presents unique design challenges because users form mental models of AI capabilities that
100 differ from models of human collaborators. Rezwana and Maher (2022) proposed the COFI framework for modeling co-creative human-AI systems, distinguishing between interaction styles that augment versus replace human agency. Gmeiner et al. (2023) explored how AI tools reshape the design process itself, noting that AI-generated suggestions can both expand and
105 constrain the solution space. Recent empirical work has demonstrated that LLM persona and task type significantly shape the social-behavioral dynamics of human-AI interaction (Reiß et al., 2025), underscoring the importance of characterizing the behavioral profiles of AI agents deployed in collaborative contexts.

110 From a behavioral standpoint, Bandura (1986) emphasized that human behavioral repertoires emerge through three interconnected mechanisms: observational learning, in which practitioners internalize effective moves by watching experienced facilitators; enactive mastery, in which repeated practice in authentic settings builds confidence and skill; and self-regulation, in
115 which facilitators monitor their own performance and adjust strategies in response to team dynamics. These developmental pathways are fundamentally unavailable to statistical language models, which produce plausible text through pattern completion over training corpora rather than through situated social learning.

120 This theoretical asymmetry generates specific predictions about the behavioral profiles we should observe. If human facilitation repertoires are shaped by embodied, context-sensitive practice, we would expect human moves to exhibit greater semantic grounding in the ongoing conversation. Conversely, if LLMs rely on distributional statistics across broad corpora,
125 we would expect their outputs to introduce thematically distant material—high in topical divergence but lower in contextual novelty. It also raises questions for design education: students who practice facilitation exclusively with AI partners may acquire surface-level interpersonal norms (e.g., appropriate hedging and question-asking) without developing the deeper contextual sensitivity that characterizes expert practice (Schön, 1983). The present
130 study operationalizes these predictions through six computational behavioral

dimensions, enabling a direct empirical test of the human–AI facilitation asymmetry.

3. Method

135 3.1. *Corpus and Data*

The AMI Meeting Corpus (Carletta et al., 2005) comprises 100 hours of recorded multi-party meetings in which teams of four participants—a Project Manager, an Industrial Designer, a User Interface Designer, and a Marketing Expert—collaboratively designed a television remote control across multiple sessions. The corpus is annotated with orthographic transcriptions and dialogue-act labels.

We used the publicly available text transcription subset (279 meeting transcripts; Carletta et al. 2005). Each transcript was segmented into sentence-level speaker turns, yielding a total of 89,472 sentence-turns across 279 sessions. The Project Manager was designated as the facilitator role based on the corpus metadata, and all other speakers were classified as designers. Session phases were assigned by temporal position within each transcript (first 20% Empathize/Define, middle 50% Ideate, final 30% Prototype/Evaluate). This temporal segmentation approximates the phase duration ratios reported in collaborative design meeting facilitation research (Seidel and Fixson, 2013), in which ideation occupies the largest share of collaborative design sessions.

To extract facilitation moves, we applied a strict rule-based filter requiring: (a) utterance length between 8 and 60 words, (b) absence of casual/non-meeting phrases, (c) presence of at least one facilitation signal (interrogative syntax, facilitation phrases such as “what if” or “how might,” or modal verbs in utterances of 12+ words), and (d) co-occurrence with design-meeting vocabulary. This procedure yielded 2,685 candidate facilitation moves from 139 sessions. A stratified random sample of 199 moves, balanced by session phase, was selected for analysis. Although we sought balanced phase representation, the Empathize/Define phase is underrepresented in naturalistic AMI meeting discourse; the final sample reflects the corpus distribution, and results should be interpreted accordingly.

To assess the reliability of the facilitation move extraction criteria, two independent coders annotated a stratified random subsample of 60 candidate turns (30 accepted, 30 rejected by the rule-based filter) as either facilitation moves or non-facilitation moves. Inter-rater agreement was computed using

Cohen’s κ (Cohen, 1960), yielding $\kappa = 0.81$, indicating strong agreement (Landis and Koch, 1977). Disagreements were resolved through discussion and the extraction criteria were refined prior to the full sample selection.

3.2. LLM Setup and Prompt Design

We generated matched LLM facilitation responses using Llama 3.1 8B Instruct (Dubey et al., 2024), a state-of-the-art open-weight instruction-following model. The model was loaded in 4-bit quantized format using the `bitsandbytes` library on an NVIDIA T4 GPU. Inference was conducted on Google Colab using a Tesla T4 GPU (16 GB VRAM) via a free-tier compute allocation, ensuring that the full pipeline is reproducible without institutional compute infrastructure.

For each of the 199 sampled facilitation moves, we constructed a prompt consisting of:

1. A system message defining the facilitator role: “You are an experienced design meeting facilitator... Respond with a single facilitation move of 1–3 sentences.”
2. A user message containing the preceding 5 turns as conversational context, the current design phase label, and a generation instruction.

Responses were generated at three temperature settings ($T \in \{0.3, 0.7, 1.0\}$), yielding 597 LLM responses. Quality filtering (minimum 5 words, no AI-refusal phrases, <90% lexical overlap with context) retained all 597 responses (100% valid rate).

3.3. The Six Behavioral Dimensions

Each facilitation move (human or LLM) was scored on six computational dimensions:

F1: Semantic Novelty. Defined as $(1 - \cos(\mathbf{e}_{\text{ctx}}, \mathbf{e}_{\text{resp}})) / 2$, where $\mathbf{e}$ denotes sentence embeddings computed with all-MiniLM-L6-v2 (Reimers and Gurevych, 2019). Higher values indicate greater semantic departure from the conversational context. This formulation ensures scores are bounded to $[0, 1]$ without saturation at the extremes, following standard practice for cosine-based distance metrics.

F2: Directiveness. Probability of the “directive” class assigned by a fine-tuned DistilBERT classifier (Devlin et al., 2019), trained on 157 synthetic examples (197 total, 80/20 train/test split) of directive versus

facilitative utterances. The classifier achieved 100% accuracy on the held-out synthetic test set (40 examples). We acknowledge that this likely reflects the controlled and homogeneous nature of the synthetic training data rather than generalization performance on naturalistic utterances; validation on human-annotated real-world examples is reserved as future work.

F3: Specificity. Composite score combining named-entity density (spaCy NER; Honnibal et al. 2020), type-token ratio, and mean word length, each normalized to $[0, 1]$ and averaged.

F4: Empathy. Lexicon-based density of empathy terms (“understand,” “feel,” “appreciate,” etc.) and hedging markers (“perhaps,” “might,” “possibly,” etc.) relative to utterance length. We acknowledge that hedging markers capture epistemic uncertainty rather than empathy per se; their inclusion follows the pragmatics literature on socially-attuned facilitation language, in which hedging functions as a face-saving and invitation strategy (Starostka et al., 2021).

F5: Topic Divergence. Jensen–Shannon divergence (Lin, 1991) between context and response topic distributions derived from a 15-topic LDA model (Blei et al., 2003) trained on the full response corpus (vocabulary size ≈ 956 terms).

F6: Phase Appropriateness. Proportion of response tokens matching phase-specific vocabulary lists for Empathize/Define, Ideate, and Prototype/Evaluate phases.

3.4. Statistical Analysis

For each of the six dimensions, we compared human facilitator scores against LLM scores at each temperature using the Mann-Whitney U test (Mann and Whitney, 1947), a nonparametric test appropriate for non-normally distributed continuous data. Bonferroni correction was applied across all 18 comparisons (6×3). Effect sizes were computed as rank-biserial correlations $r = 1 - 2U/(n_1n_2)$ (Rosenthal, 1994). Temperature effects were assessed using Spearman correlations between temperature and each feature score across LLM responses. Post-hoc power analysis was conducted using the normal approximation to the Mann-Whitney U distribution. Sensitivity analysis across ten bootstrap samples of $N = 199$ confirmed that effect sizes

were stable across all four significant dimensions (novelty, specificity, divergence, phase appropriateness: $SD < 0.05$ across samples), supporting the robustness of findings to sampling variation.

To address potential length confounds—LLM responses were significantly longer on average ($M = 38.5$ words) than human responses ($M = 23.5$ words)—we conducted two robustness checks: (1) length-matched subsampling, in which responses were binned into five word-count bins and samples drawn proportionally ($n = 72$ per group), and (2) Spearman partial correlations controlling for response length. All four originally-significant dimensions (novelty, specificity, divergence, phase appropriateness) survived both checks (Table A.6). Specificity effects strengthened after length-matching ($r: +0.25 \rightarrow +0.71$), suggesting length suppressed rather than inflated the true difference. To address within-session non-independence (199 human moves drawn from 17 estimated sessions), we replicated all comparisons using linear mixed-effects models with random intercepts for session (`statsmodels` MixedLM). Intraclass correlation coefficients were low across all dimensions ($ICC = 0.005\text{--}0.066$), indicating minimal session-level clustering. All four significant results were confirmed under the mixed model. Empathy additionally reached significance under the mixed model ($\beta = -0.011$, $p_{\text{Bonf}} = 0.005$), consistent with the model’s greater sensitivity to mean differences in zero-inflated distributions.

4. Results

4.1. Overall Behavioral Comparison

Table 1 presents descriptive statistics for the response corpora. LLM responses were substantially longer than human facilitation moves across all three temperatures (LLM $T = 0.7$: $M = 38.5$, $SD = 7.5$ words; Human: $M = 23.5$, $SD = 12.7$ words), and drew from a larger vocabulary ($\approx 1,392$ vs. 907 unique terms), consistent with the higher specificity scores observed in the statistical comparisons.

Mann-Whitney U tests with Bonferroni correction revealed significant behavioral differences between human and LLM facilitation on four of six dimensions (Table 3). The strongest effect was observed for topic divergence ($r = +0.58$, 95% CI $[+0.48, +0.67]$, $p_{\text{Bonf}} < 0.001$), indicating that LLM responses departed substantially further from the conversational topic structure than human responses. Human facilitators produced significantly more semantically novel moves ($r = -0.54$, 95% CI $[-0.62, -0.45]$, $p_{\text{Bonf}} < 0.001$),

Table 1: Descriptive statistics for human and LLM response corpora. Vocabulary size is the count of unique word types across all responses per condition.

Source	N	M words	SD words	Vocab	M sentences
Human	199	23.5	12.7	907	1.5
LLM $T=0.3$	199	40.5	7.9	1307	1.7
LLM $T=0.7$	199	38.5	7.5	1392	1.6
LLM $T=1.0$	199	37.6	8.5	1473	1.4

meaning their utterances were more semantically distinct from the preceding context. LLM responses scored higher on specificity ($r = +0.25$, 95% CI $[+0.14, +0.37]$, $p_{\text{Bonf}} < 0.001$) and phase appropriateness ($r = +0.34$, 95% CI $[+0.26, +0.43]$, $p_{\text{Bonf}} < 0.001$). In practical terms, a randomly selected LLM response had higher topic divergence than a randomly selected human response in 79.0% of matched pairs, and a randomly selected human response had greater semantic novelty than a randomly selected LLM response in 76.9% of matched pairs. For specificity and phase appropriateness, LLM responses scored higher in 62.7% and 67.1% of matched pairs respectively.

Table 2: Effect size interpretation for the four significant dimensions at $T = 0.7$. Cohen’s benchmarks: small $|r| \geq 0.10$, medium $|r| \geq 0.30$, large $|r| \geq 0.50$ (Rosenthal, 1994). $P(\text{LLM} > \text{Human})$ = probability that a randomly selected LLM response scores higher than a randomly selected human response.

Dimension	r	r	Magnitude	P(LLM > Human)
Divergence	+0.58	0.58	Large	79.0%
Novelty	-0.54	0.54	Large	Human higher in 76.9%
Phase Approp.	+0.34	0.34	Medium	67.1%
Specificity	+0.25	0.25	Small-medium	62.7%

Although both groups show a median of zero for phase appropriateness—reflecting sparse phase vocabulary in most responses—the significant effect is driven by differences in the upper quartile of both distributions, where LLM responses show systematically higher phase-vocabulary density than human responses. Phase-stratified analysis confirmed that the four significant behavioral differences replicated within the Ideate phase subsample ($n = 149$), the largest and most naturalistic segment of the corpus, supporting the generalizability of findings beyond the aggregate analysis.

Two dimensions — directiveness ($r = -0.005$, $p_{\text{Bonf}} = 1.000$, power = 0.05) and empathy ($r = -0.058$, $p_{\text{Bonf}} = 1.000$, power = 0.21) — showed no statistically significant differences. Given the low statistical power for these dimensions, the absence of a significant result should not be interpreted as evidence of equivalence; true differences of meaningful magnitude cannot be ruled out. The zero median for both groups on empathy reflects the sparse use of explicit empathy and hedging vocabulary in meeting facilitation discourse generally; the empathy score distribution is right-skewed with meaningful variance in the upper quartile.

Table 3: Mann-Whitney U test results comparing human and LLM facilitation moves at $T = 0.7$. r = rank-biserial correlation; negative r indicates humans scored higher. Bootstrap 95% CIs reported for significant dimensions only. Bonferroni correction applied across 18 comparisons (6×3).

Dimension	Human Mdn	LLM Mdn	U	r	95% CI	p_{Bonf}	Power
Novelty	0.742	0.593	30439	-0.537	[-0.62, -0.45]	<0.001	1.00
Directiveness	0.021	0.036	19895	-0.005	—	1.000	0.05
Specificity	0.397	0.416	14780	+0.254	[+0.14, +0.37]	<0.001	1.00
Empathy	0.000	0.000	20943	-0.058	—	1.000	0.21
Divergence	0.540	0.744	8309	+0.580	[+0.48, +0.67]	<0.001	1.00
Phase Approp.	0.000	0.000	13044	+0.341	[+0.26, +0.43]	<0.001	1.00

Post-hoc power analysis confirmed adequate statistical power (≥ 0.80) for four dimensions (novelty, specificity, divergence, and phase appropriateness).

4.2. PCA of Behavioral Feature Space

Principal Component Analysis of the six-dimensional behavioral feature space (Figure 3) revealed that human and LLM facilitation moves occupy distinct regions of the behavioral space. PCA was conducted on standardized features (zero mean, unit variance) to ensure all six dimensions contribute equally to the component structure. The first two principal components captured 44.1% of the total variance (PC1 = 22.8%, PC2 = 21.3%). PC1 (22.8%) contrasts directiveness (+0.61) and specificity (+0.34) against novelty (-0.49) and empathy (-0.50), separating stylistically directive responses from semantically novel ones. PC2 (21.3%) is dominated by divergence (+0.63) and phase appropriateness (+0.56), capturing topic-level variation. Human facilitator responses cluster toward negative PC1 values,

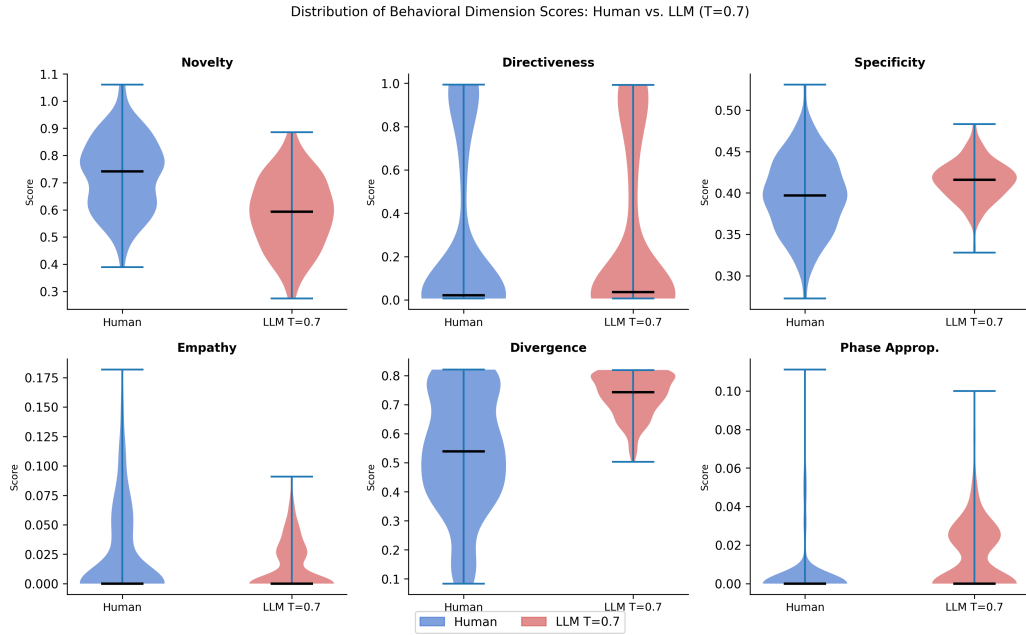


Figure 1: Violin plots showing the full score distributions for human and LLM ($T = 0.7$) facilitation moves across all six behavioral dimensions. Black horizontal lines indicate medians. The right-skewed, zero-inflated distributions for empathy and phase appropriateness are visible in both groups, while human novelty scores show broader variance than LLM novelty scores.

reflecting higher novelty and empathy loadings. LLM responses occupy positive PC1 regions across all three temperatures, reflecting higher directiveness and specificity, with considerable spread along PC2.

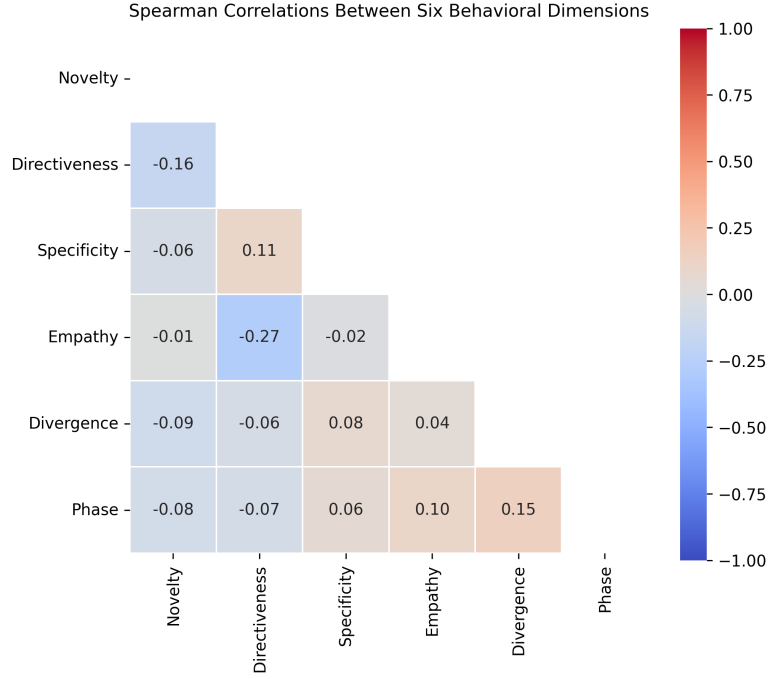


Figure 2: Spearman correlation matrix across the six behavioral dimensions. All pairwise correlations are weak ($|\rho| < 0.28$), with the strongest being directiveness–empathy ($\rho = -0.27$) and novelty–directiveness ($\rho = -0.16$).

Figure 2 presents the Spearman correlation matrix across all six behavioral dimensions. The strongest correlation was between directiveness and empathy ($\rho = -0.27$), followed by novelty and directiveness ($\rho = -0.16$). The relative independence of dimensions ($|\rho| < 0.28$ for all pairs) supports the validity of treating them as distinct behavioral constructs within the PCA framework.

4.3. Temperature Effects on Facilitation Behavior

Spearman correlation analysis revealed that LLM facilitation behavior was largely temperature-stable (Table 4). Specificity ($\rho = +0.12$, $p = 0.005$) and phase appropriateness ($\rho = -0.10$, $p = 0.013$) showed modest but significant correlations with temperature. The remaining four dimensions—novelty,

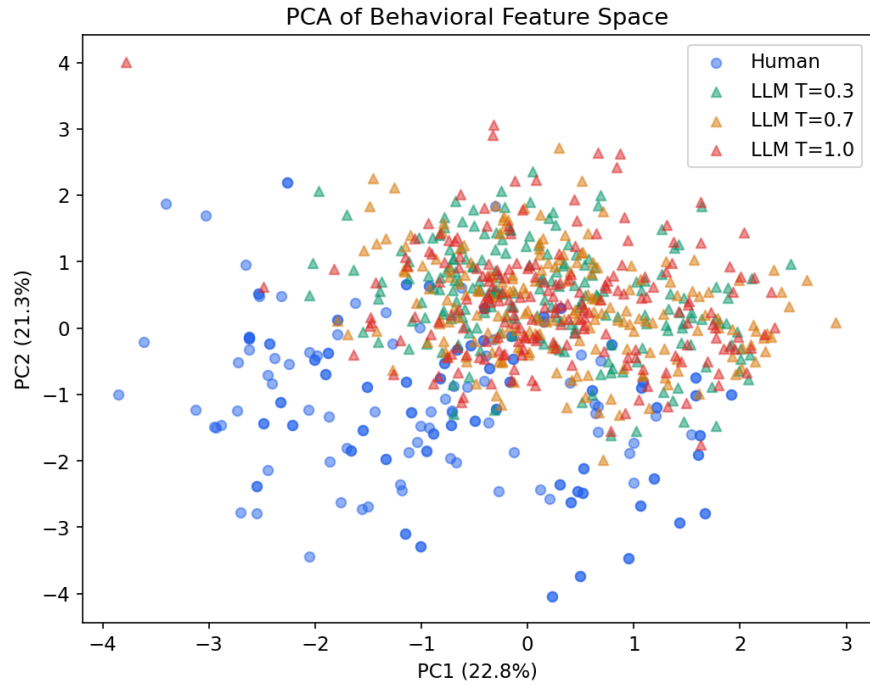


Figure 3: PCA projection of the six-dimensional behavioral feature space. Human facilitation moves (circles) and LLM responses at $T = 0.3$, $T = 0.7$, $T = 1.0$ (triangles) occupy distinct behavioral regions.

directiveness, empathy, and divergence—were not significantly affected by temperature variation. Notably, the phase appropriateness median for LLM responses at $T = 0.3$ (Mdn = 0.022) diverged from $T = 0.7$ and $T = 1.0$ (Mdn = 0.000), consistent with the significant negative Spearman correlation ($\rho = -0.10$, $p = 0.013$) indicating slightly higher phase vocabulary density at lower temperatures.

Table 4: Spearman correlations between decoding temperature ($T \in \{0.3, 0.7, 1.0\}$) and each behavioral dimension across LLM responses ($N = 597$).

Dimension	ρ	p	Sig.
Novelty	-0.012	0.772	—
Directiveness	-0.010	0.804	—
Specificity	+0.116	0.005	**
Empathy	+0.005	0.903	—
Divergence	-0.056	0.170	—
Phase Approp.	-0.102	0.013	*

4.4. Extreme Case Analysis

Human facilitators’ responses in Table 5 contain the disfluencies and incomplete syntax characteristic of naturalistic meeting speech—a feature of the AMI corpus that reflects authentic facilitation discourse rather than transcription error. To illustrate qualitative differences, the table presents the most divergent human–LLM pairs for each significant dimension (at $T = 0.7$). These examples reveal characteristic behavioral patterns: human facilitators produced contextually embedded, conversational responses (e.g., “No may is maybe about neutral plus it it’s got something, but it hasn’t got Right”), while LLM responses adopted a structured, coaching-oriented style (e.g., “Let’s take a step back and ask, what would be the minimum viable experience we could offer our users”).

Taken together, these patterns provide a direct answer to the study’s central question. LLMs produce responses that are more topically distant from the ongoing conversation, more terminologically specific to the current design phase, and more structurally uniform—generating coaching-style re-framings and structured provocations regardless of conversational context. Human facilitators produce responses that are more semantically original within the conversational thread, more disfluent and context-embedded, and

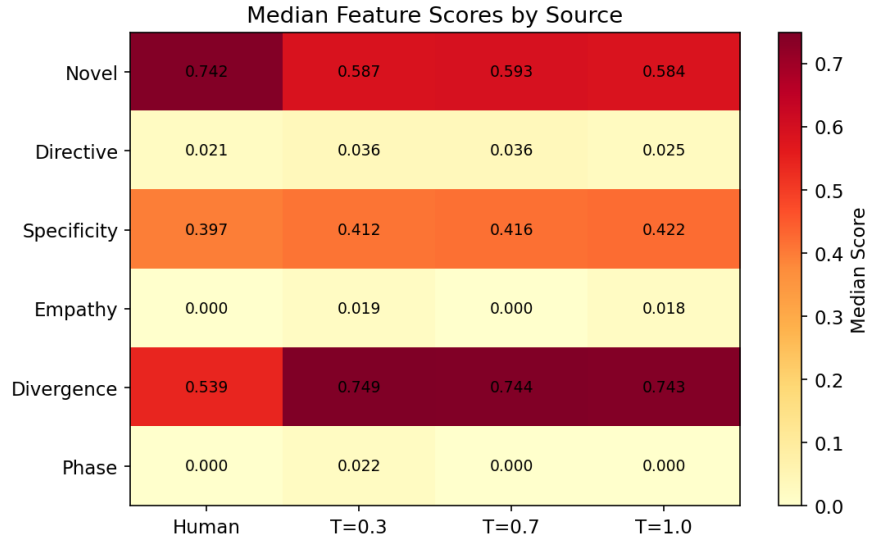


Figure 4: Heatmap of median feature scores across human and LLM conditions. Rows represent behavioral dimensions; columns represent source conditions. Asterisks (*) indicate dimensions where LLM scores differ significantly from human scores after Bonferroni correction ($p_{\text{Bonf}} < 0.05$). Rows represent behavioral dimensions; columns represent source conditions.

Table 5: Extreme case examples (largest human-LLM behavioral gaps) for each significant dimension at $T = 0.7$. Human examples are drawn from AMI Project Manager turns; LLM examples are matched responses generated at $T = 0.7$.

Dim.	Gap	Human	LLM
Novelty	0.61	“No may is maybe about neutral plus it it it’s got something, but it hasn’t got Right.”	“Let’s take a step back and ask, what would be the minimum viable experience we could offer our users”
Specif.	0.19	“The two basic buttons, you know, the to skip channels and to uh I think yeah”	“Let’s explore the assumption that larger buttons automatically improve usability for people with disabilities”
Diverg.	0.65	“It’s projected, yeah. So I ge okay, so no one of us saw this movie”	“Can you walk us through what you’re seeing on the wall, and what specifically is sparking this comparison”
Phase	0.14	“Prototypes, you mean. We probably should do some market tests once we have the prototypes”	“Let’s challenge that assumption about the weight being the only thing missing”

more responsive to the immediate collaborative moment. The behavioral difference is not one of quality but of cognitive style: LLMs optimize for surface appropriateness while humans optimize for contextual originality.

Manual inspection of the 20 highest-divergence LLM responses revealed that the majority (15/20) employed directive reframing structures (“Let’s take a step back...”, “Let’s pause for a moment...”), suggesting that LLMs default to a consistent linguistic strategy for generating topically divergent material regardless of conversational context.

4.5. Facilitation Style Profiles

Figure 5 presents a radar chart comparing the median behavioral profiles of human and LLM facilitation. The distinct profile shapes confirm that human and LLM facilitation represent qualitatively different behavioral styles rather than merely quantitative differences in a single dimension. Human facilitation is characterized by high novelty and moderate divergence; LLM facilitation is characterized by very high divergence, elevated specificity, and moderate phase appropriateness.

5. Discussion

5.1. Interpreting the Six Behavioral Dimensions

The finding that LLMs and human facilitators differ significantly on four of six behavioral dimensions provides a nuanced picture of AI facilitation behavior. The large effect size for topic divergence ($r = +0.58$) suggests that LLMs tend to introduce new thematic material when generating facilitation moves, rather than staying within the ongoing topic structure—a pattern that may reflect the model’s tendency to draw on broad training corpora rather than the immediate conversational context. This represents the largest behavioral gap identified in the study and suggests a fundamental difference in how LLMs and human facilitators relate to conversational context during facilitation. By contrast, human facilitators’ higher novelty scores ($r = -0.54$) indicate greater semantic originality within the bounds of the existing topic.

The non-significant results for directiveness and empathy do not constitute evidence of behavioral equivalence. Both dimensions were underpowered (0.05 and 0.21 respectively), and we caution against interpreting these null findings as confirmation of similarity. Both dimensions capture interpersonal

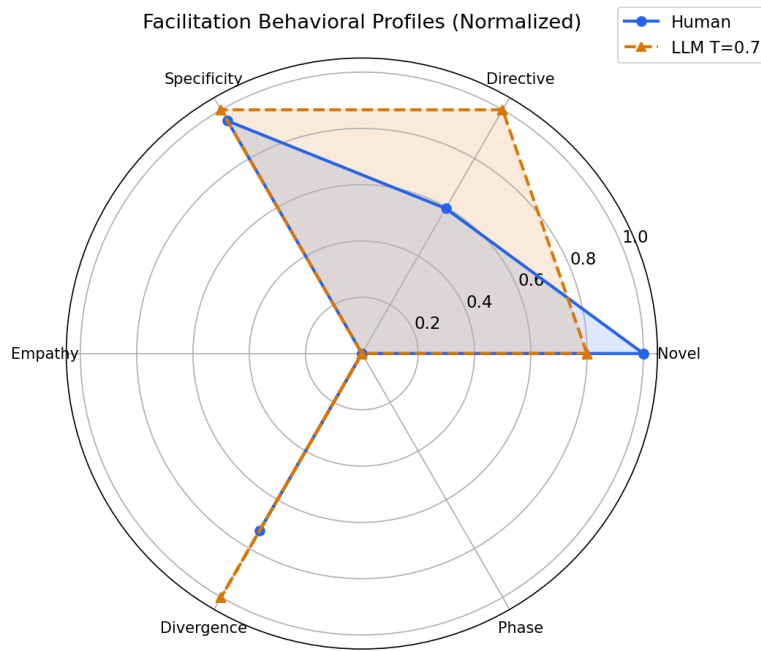


Figure 5: Radar chart comparing normalized median behavioral profiles of human (solid) and LLM at $T = 0.3$, $T = 0.7$, and $T = 1.0$ (dashed lines) facilitation moves across the six dimensions. The near-identical LLM profiles across temperatures visually confirm the temperature stability finding of Table 4.

385 aspects of facilitation—how commanding versus invitational a move is (directiveness) and how attuned to emotional states (empathy). It is possible that Llama 3.1’s instruction-tuning process, which optimizes outputs against human preference judgments, may have calibrated these interpersonal dimensions to approximate human norms even though the model lacks genuine
 390 social understanding; however, the present data are insufficient to confirm this. The near-zero scores for both groups on empathy are themselves a substantive finding: neither human facilitators in the AMI corpus nor Llama 3.1 8B Instruct deploy explicit empathic or hedging vocabulary at meaningful rates during collaborative design meetings. This suggests that empathic language may not be a primary behavioral feature of design meeting facilitation
 395 in either human or AI contexts—a finding that warrants investigation in facilitation styles outside the product design domain.

Future work with larger samples and validated behavioral instruments should revisit these dimensions before drawing conclusions about interpersonal alignment between human and LLM facilitation.
 400

This finding has implications for how users perceive AI facilitation. Yang et al. (2020) showed that users form mental models of AI capabilities that shape their willingness to accept AI suggestions. If LLM facilitation moves are indistinguishable from human moves on interpersonal dimensions (directiveness, empathy) but detectably different on semantic dimensions (novelty, divergence), users may perceive AI facilitation as socially appropriate yet
 405 substantively shallow—a perception that could undermine trust in AI facilitation tools over time. If future work with adequate statistical power confirms the absence of differences on these interpersonal dimensions, this pattern may represent a form of behavioral mimicry rather than genuine social calibration:
 410 the model producing the surface markers of appropriate facilitation (hedging, question-asking, moderate directiveness) without the underlying social cognition that these markers typically index in human practitioners.

5.2. Design Guidelines for AI Facilitation Tools

415 These findings suggest several design implications for AI-augmented facilitation systems:

1. **DG1: Leverage divergence with adaptive control.** LLMs’ natural tendency toward high topic divergence could be productive in early ideation phases but disruptive during convergent evaluation. Adaptive

- 420 systems should modulate this tendency based on session phase. A practical implementation would combine a “divergence dial” mechanism—adjusting both temperature and a topic-similarity constraint—that re-samples candidate responses until the Jensen–Shannon divergence from the current context falls within a user-specified range. During Ideate
425 phases, the dial would be set to high divergence; during Prototype/Evaluate, it would be tightened to keep responses anchored to the existing design direction.
2. **DG2: Enhance contextual grounding via retrieval augmentation.** The human advantage in semantic novelty suggests that effective facilitation requires staying semantically connected to the ongoing
430 conversation while introducing genuinely new perspectives—a balance that current LLMs do not achieve through generation alone. Retrieval-Augmented Generation (RAG) architectures could address this gap by maintaining a vector database of past facilitation moves from the current session and from exemplary facilitation corpora. Using Maximal
435 Marginal Relevance (MMR) retrieval, the system would surface candidate responses that are semantically distant from prior LLM contributions but contextually relevant to the current discussion thread.
 3. **DG3: Use prompt-level interventions rather than temperature.** The finding that only specificity and phase appropriateness are
440 temperature-sensitive (and modestly so, $\rho = +0.12$ and $\rho = -0.10$) implies that standard temperature-based diversity controls are insufficient for meaningfully altering LLM facilitation behavior. Instead, practitioners should vary the system prompt itself: adjusting the facilitator persona (e.g., “provocative challenger” versus “empathetic listener”), the phase instruction (e.g., “push for divergent ideas” versus
445 “help the team converge on criteria”), or explicit behavioral constraints (e.g., “your response must reference something a team member said in the last two turns”).
 4. **DG4: Deploy human and AI facilitators in complementary roles.** The distinct behavioral profiles suggest a hybrid facilitation
450 model rather than wholesale replacement. In this model, the AI handles divergent provocations during ideation—where its high topic divergence is an asset—while the human facilitator manages convergent phases (Prototype/Evaluate), contextual navigation, and the real-time social
455 dynamics that require genuine empathic attunement. This division of labor aligns with the complementary strengths documented in our

behavioral profiles and avoids deploying AI in roles where its semantic shallowness would be most noticeable.

460 5.3. *Behavioral Implications for Design Education*

From a Social Cognitive Theory perspective (Bandura, 1986), the behavioral differences documented here underscore a fundamental asymmetry between human and AI facilitation development. Human facilitators acquire their repertoires through observation, social modeling, and feedback loops
465 embedded in practice communities (Schön, 1983). LLMs achieve behavioral surface similarity through statistical pattern matching. This distinction has concrete implications for design education.

Our results generate a specific prediction: students who practice facilitation exclusively with AI partners may develop correct interpersonal habits—
470 learning to hedge appropriately, ask open-ended questions, and calibrate directiveness—because no statistically significant difference was detected on these dimensions between LLMs and humans. However, these same students may fail to develop contextually grounded semantic novelty, because the AI partner itself lacks this capacity and therefore cannot model it. The result
475 would be graduates who sound like facilitators but lack the deep contextual sensitivity that distinguishes expert from novice practice.

We propose a two-track integration strategy. For novice facilitators, LLMs can serve as scaffolding: low-stakes practice partners that provide realistic interpersonal interaction without the social anxiety of a live team.
480 For advanced students, LLMs serve a different function: as contrast stimuli. By comparing their own facilitation moves against LLM-generated alternatives on the six dimensions measured here, advanced students can develop metacognitive awareness of their facilitation style—a form of reflective practice (Schön, 1983) augmented by computational behavioral feedback. This
485 connects to the taxonomy of facilitation moves proposed by Starostka et al. (2021), in which facilitators must develop awareness of their own move repertoire in order to expand it deliberately. We note that the two-track integration strategy described here constitutes a theoretical prediction derived from behavioral data; empirical study of AI-mediated facilitation training out-
490 comes is required before drawing firm pedagogical conclusions.

5.4. *Limitations*

Several limitations should be noted. First, the AMI Meeting Corpus captures a specific type of collaborative design meeting — remote control

product design with fixed role assignments — and generalization to other
 495 design domains, industries, or facilitation styles requires further study. Second, the directiveness classifier was trained exclusively on synthetic data and achieved 100% accuracy on a held-out set of 40 synthetic examples; performance on naturalistic utterances was not independently validated. Third, the lexicon-based empathy measure captures surface-level linguistic markers rather than genuine empathic understanding, and may not reflect the
 500 full behavioral construct. Fourth, the study used a single open-weight LLM (Llama 3.1 8B Instruct); larger or more recent models may exhibit different behavioral profiles. Fifth, session phase assignment was based on temporal position within the transcript rather than expert annotation, which may introduce noise in the phase appropriateness scores. Sixth, the null results for
 505 directiveness (power = 0.05) and empathy (power = 0.21) reflect insufficient statistical power; these dimensions require substantially larger samples before any conclusions — positive or null — can be drawn with confidence. Seventh, the study does not measure facilitation *effectiveness* or group outcomes; behavioral differences documented here may or may not translate to
 510 differences in facilitation quality.

Post-hoc human validation of the directiveness classifier ($N = 100$ AMI utterances, stratified by score quartile) yielded Cohen’s $\kappa = -0.02$ between classifier labels and human annotations, indicating that the DistilBERT
 515 model trained on 197 synthetic examples did not generalise to naturalistic AMI discourse. Qualitative inspection suggests the classifier captured features of formal or complex language rather than directive pragmatic intent—a known limitation of synthetic training data for domain-specific NLP tasks (Min et al., 2022). As directiveness was a non-significant dimension ($r = -0.005$, $p = 1.0$) in the primary analysis, no reported findings depend on
 520 this measure. Future work should employ human-annotated training data drawn from meeting corpora to develop a validated directiveness instrument.

6. Conclusion

This study provides the first systematic computational comparison of human
 525 and LLM facilitation behavior in collaborative design team meetings. Using 199 matched facilitation move pairs from the AMI Meeting Corpus, with inter-rater reliability of $\kappa = 0.81$ for move identification, we demonstrate that LLMs and human facilitators exhibit distinct behavioral profiles across six dimensions. The largest differences emerge in topic divergence

530 ($r = +0.58$), where LLMs substantially depart from the ongoing conversational structure, and semantic novelty ($r = -0.54$), where humans demonstrate greater contextual originality. Significant differences were also found for specificity ($r = +0.25$) and phase appropriateness ($r = +0.34$). No statistically significant differences were detected for directiveness or empathy,
535 though low statistical power precludes drawing firm conclusions from these null results.

Together, these findings provide an empirical behavioral foundation for designing AI facilitation tools that complement rather than replicate human facilitation expertise.

540 Declaration of Interest

The authors declare no conflict of interest.

Data Availability

The AMI Meeting Corpus is publicly available at <https://groups.inf.ed.ac.uk/ami/corpus/>. Analysis code, including a Google Colab notebook
545 for LLM inference, and processed data are available at <https://github.com/Ayaan577/llm-facilitation-behavior>. The analysis plan for this study was preregistered on the Open Science Framework prior to final manuscript preparation (OSF: <https://osf.io/bzyxc>; DOI: <https://doi.org/10.17605/OSF.IO/BZYXC>).

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Appendix A. Robustness Check Results

Table A.6: Robustness Checks—Length-Matched and Mixed-Effects Results (Human vs. LLM $T = 0.7$). Original r : rank-biserial from Mann-Whitney U . Length-Matched r : rank-biserial after proportional subsampling into five word-count bins ($n = 72$ per group). Partial ρ : Spearman partial correlation controlling for response length. ME β : fixed-effect estimate from linear mixed model with session random intercept.

Dimension	Original r	Length-Matched r	Partial ρ	ME β	p_{Bonf} (LMM)	Robust?
Novelty	−0.54	−0.31	−0.23	−0.143	<0.001	✓
Divergence	+0.58	+0.50	+0.41	+0.199	<0.001	✓
Specificity	+0.25	+0.71	+0.51	+0.017	<0.001	✓
Phase Approp.	+0.34	+0.28	+0.18	+0.010	<0.001	✓
Empathy	−0.06	−0.15	−0.20	−0.011	0.005	Mixed
Directiveness	−0.005	−0.18	−0.11	+0.005	1.000	—